

YIHANG CHEN

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EDUCATION

UC San Diego, La Jolla, CA

Sep 2025 – (Expected) Jun 2027

M.S. in Computer Science and Engineering, **GPA:** 3.96 / 4.0

Zhejiang University, Hangzhou, China

Sep 2021 – Jun 2025

B.Eng. in Computer Science, **GPA:** 3.97 / 4.0

SKILLS

- **Programming Languages:** Python, Golang, Kotlin, TypeScript, SQL
- **AI & Machine Learning:** PyTorch, HuggingFace, NLP, LLM Agents, Speech Processing
- **Web Frontend & Mobile:** Vue3, Android Native (Jetpack Compose, Material 3)
- **Web Backend & Tools:** Python FastAPI, Golang Gin, Linux, Docker, Git

EXPERIENCE

Google, Shanghai, China

Jun 2026 – Aug 2026

Software Engineering Intern

- Developed a feature for a Pixel system app using Jetpack Compose and MVVM architecture, enabling real-time rendering of complex dynamic visual content.
- Designed two-stage GenAI pipelines for flexible, automatic layout design, optimizing latency and token payload via placeholder injection while guaranteeing privacy with Private Cloud Compute.
- Implemented a secure, sandboxed export pipeline to convert dynamic UI into high-fidelity image assets for cross-app sharing.

Tencent AI Lab, Shenzhen, China

Feb 2025 – Jul 2025

Research Intern

- Researched and trained a 300M+ parameter speech generation model “F5E-TTS” on 16-GPU clusters using HuggingFace Accelerate, achieving a 16.4% relative word error rate (WER) reduction over the baseline.
- Designed multimodal alignment strategies to integrate text, semantic, and mel-spectrogram inputs into a unified conditional flow matching (CFM) framework.
- Implemented classifier-free guidance (CFG) supporting both text-to-speech and voice conversion tasks, enabling fine-grained control over multi-condition inputs.

UMass Amherst, Amherst, MA

Oct 2023 – Oct 2024

Research Assistant

- Engineered an automated multimodal data pipeline to crawl and process 600+ hours of raw videos (YouTube, RedNote) into high-quality datasets for two publications (*TalkCuts* & *RapVerse*).
- Integrated PySceneDetect (scene cuts), RTMDet/DWPose (human & pose detection), Whisper/Spleeter (speech processing), and Qwen2.5-VL (video captioning) to automate data sanitization and alignment.
- Evaluated baseline models on our custom dataset to establish quantitative benchmarks.

PUBLICATIONS

- **ICASSP 2026** (First Author) – *F5E-TTS: Enhancing Speech Synthesis by Aligning Text with Rich Semantic Representations*
- **NeurIPS D&B Track 2025** (Co-author) – *TalkCuts: A Large-Scale Dataset for Multi-Shot Human Speech Video Generation*
- **ICCV 2025** (Co-author) – *RapVerse: Coherent Vocals and Whole-Body Motions Generations from Text*